OpenClaw RGB Light Control Tutorial

> Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on RGB-related content.

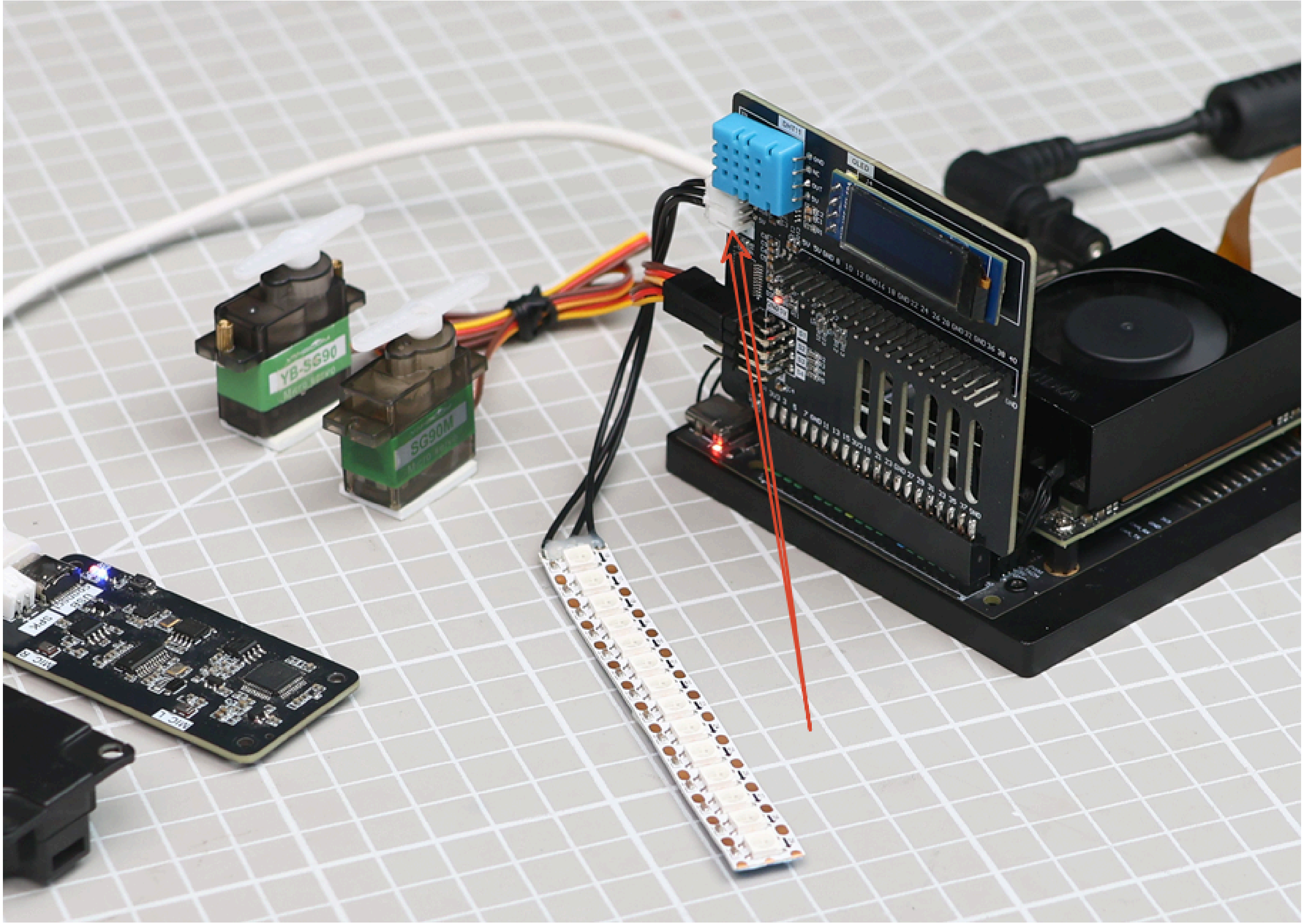
1. Tutorial Objectives

This article demonstrates OpenClaw's natural language control capabilities through RGB light peripherals. After completing this tutorial, readers can:

- Understand the basic approach to controlling RGB lights with OpenClaw
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code
- Complete four RGB cases: single color lighting, breathing light, warning flash, rainbow marquee
- Understand the complete chain from "user input" to "light execution"

2. Hardware Preparation

- OpenClaw mainboard
- RGB light strip / RGB light bead module `[Default 14 WS2812]`
- `[Optional] Microphone module`
- Wiring diagram



You can add RGB wiring diagram and hardware physical diagram here.

3. Software Preparation

Code entry points that will be used in the tutorial:

- Low-level driver code: `/home/jetson/hardware_hal_demo/`
- RGB control command: `/home/jetson/hardware_hal_demo/rgb`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

The `rgb` tool already supports the following capabilities:

- Set fixed color `color`
- Set light effect `effect`
- Turn off lights `off`

Current common light effects include:

- `breathe` Breathing light
- `blink` Flash
- `flow` Flowing light
- `marquee` Marquee
- `rainbow` Rainbow light

If you want to verify underlying RGB commands individually, you can also input the following in the terminal:

```
rgb color --color red  #All turn red
rgb effect --effect breathe --color blue --speed 5   #Breathing light
rgb off   #Turn off lights
```

More commands can be found in the md document in this folder,

```
/home/jetson/hardware_hal_demo/README.md
```

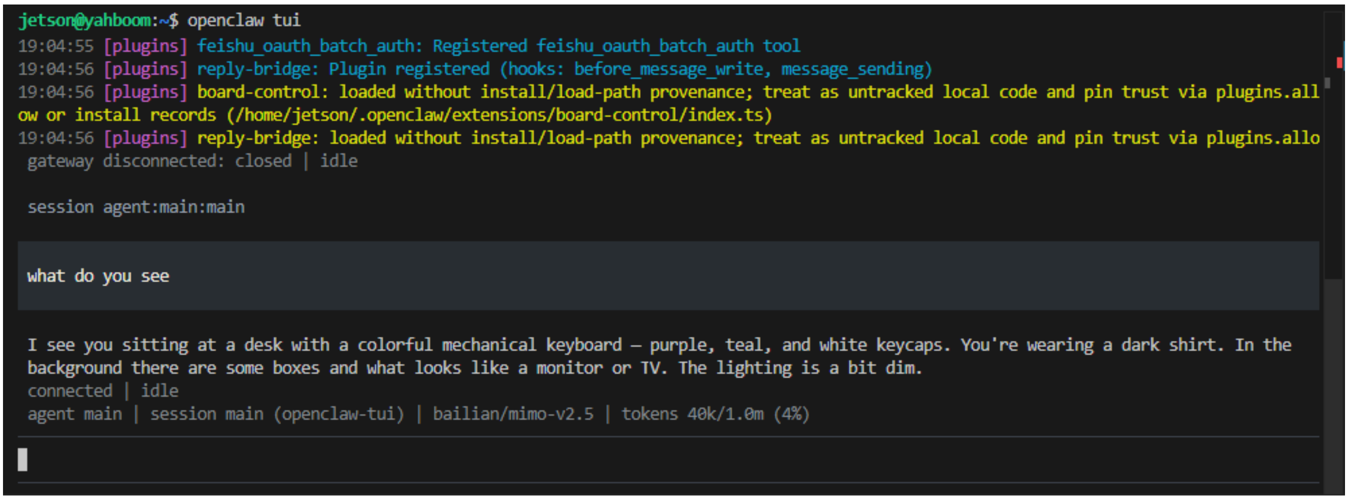
If underlying commands can execute normally, it indicates hardware and Jetson mainboard connection is normal and communication is normal.

4. Test RGB Light

Before starting this tutorial, please complete one `openclaw tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After confirming OpenClaw can respond normally, enter RGB-specific testing.

Terminal input to enter TUI:

```
openclaw tui
```



You can try inputting the following content in the terminal:

- Set all RGB lights to red
- Change RGB lights to blue
- Lower RGB light brightness
- Turn off RGB lights

```
Set all RGB lights to red

All RGB lights set to red.

Change RGB lights to blue

All RGB lights changed to blue.

Turn off RGB lights

RGB lights turned off. ●
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
44k/1.0m (4%)
```

If you want to test a specific light bead, you can also try inputting content like:

- Set the 1st light bead to green
- Set the 5th light bead to purple

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote control, classroom demonstrations	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First run through the control chain
Voice code	Voice interaction, complete AI experience	Closest to real conversation experience	As final demonstration solution

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

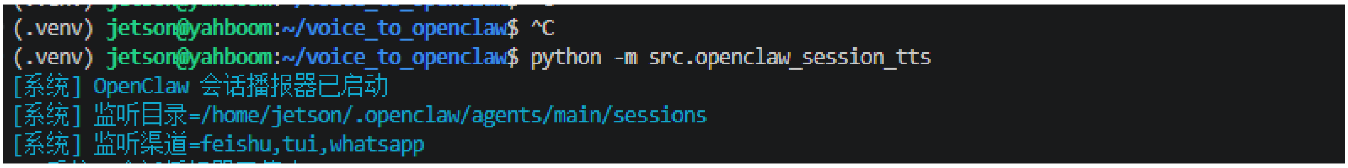
- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

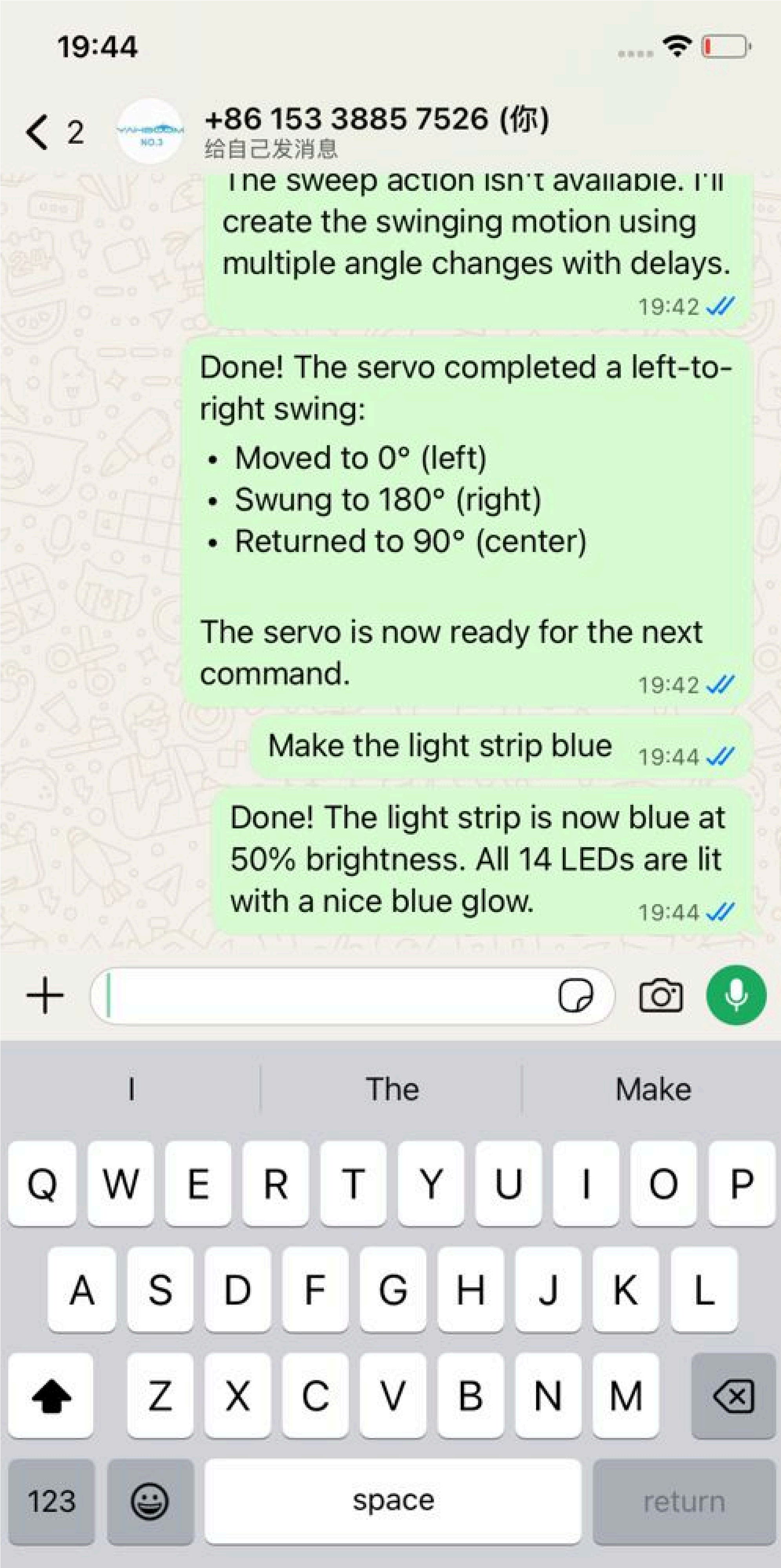


5.1 whatsapp Solution

For whatsapp connection steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. This section only retains dialogue examples suitable for the RGB tutorial.

After creation, you can directly dialogue, for example:

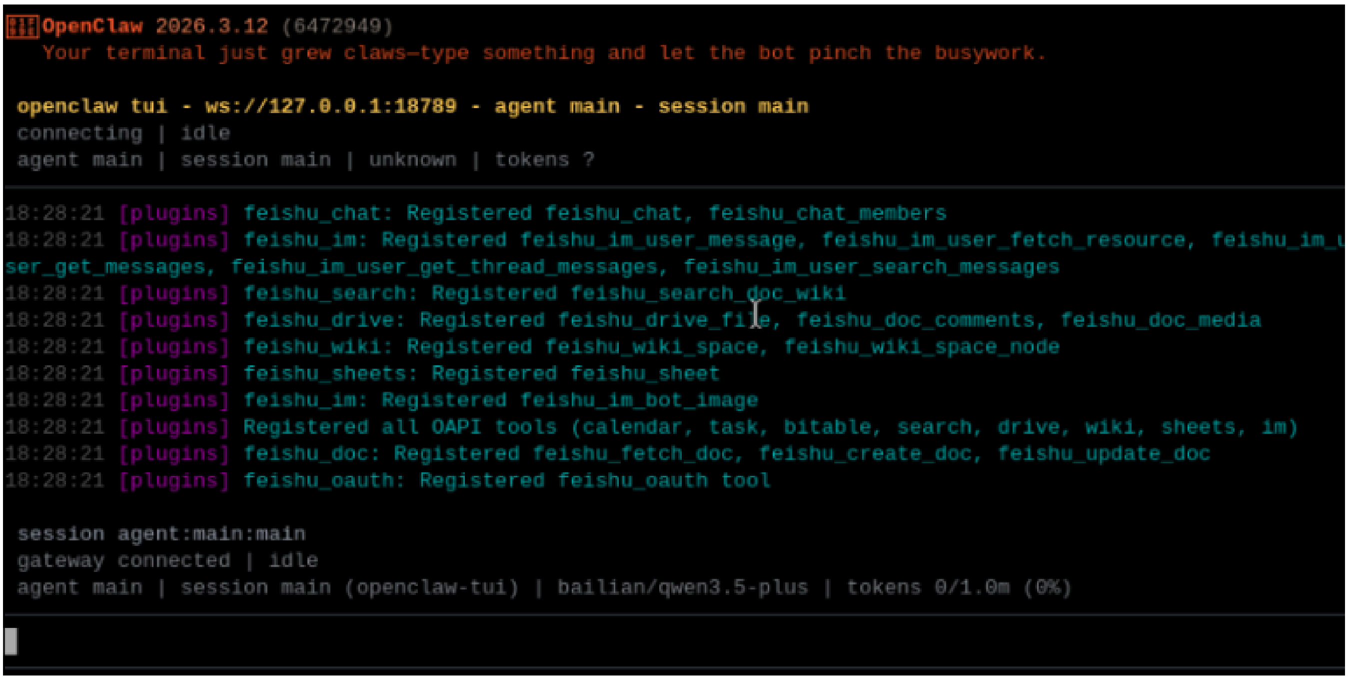
- Make the light strip blue
- Turn on rainbow light effect
- Make the lights flash faster


19:44
2
+86 153 3885 7526 (你)
给自己发消息
The sweep action isn't available. I'll create the swinging motion using multiple angle changes with delays.
19:42
Done! The servo completed a left-to-right swing:
• Moved to 0° (left)
• Swung to 180° (right)
• Returned to 90° (center)
The servo is now ready for the next command.
19:42
Make the light strip blue
19:44
Done! The light strip is now blue at 50% brightness. All 14 LEDs are lit with a nice blue glow.
19:44
I
The
Make
space
return
123

5.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter the following command in the terminal to continue the dialogue:

```
openclaw tui
```

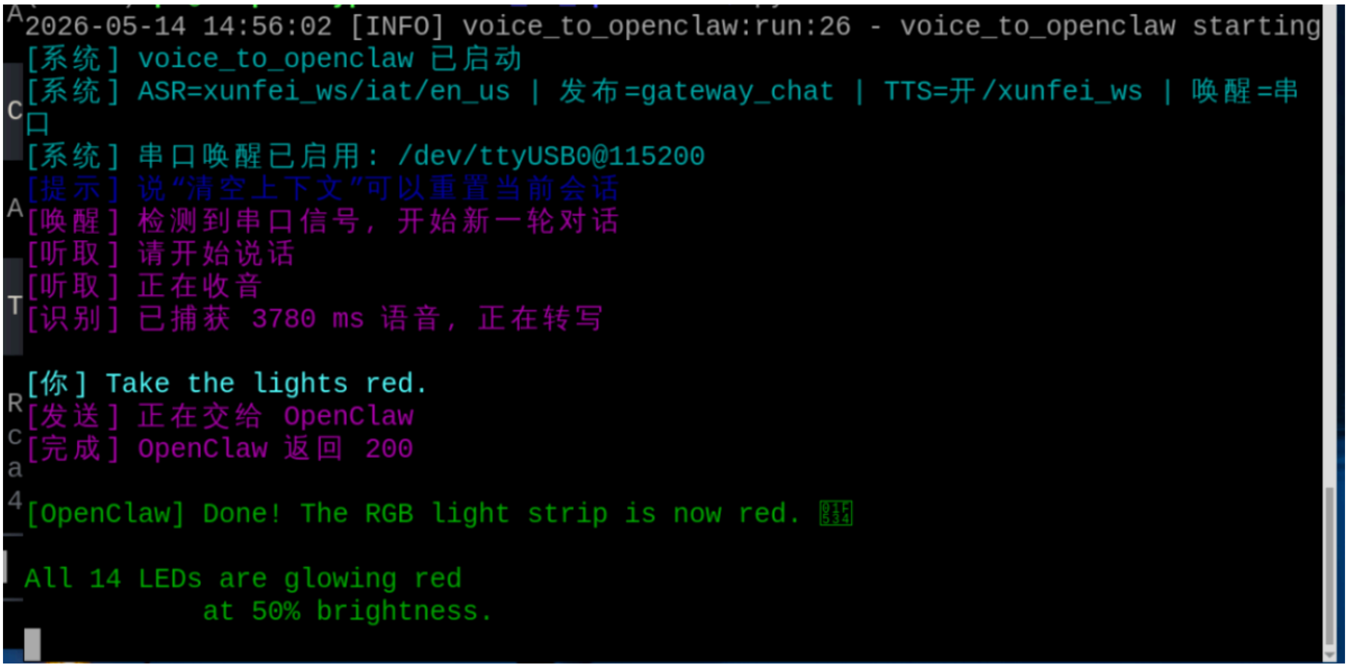


5.3 Voice Code Solution

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. After waking up, you can directly say control content, for example:

- Take the lights red
- Turn on blue breathing light
- Make the light flow like a rainbow



6. Comprehensive Cases

Note: The following are all dialogue demonstrations using `openclaw tui` method. You can also choose whatsapp or voice method to complete.

6.1 Case One: Single Color Lighting

Experimental Objective

Make RGB lights stably display a certain fixed color to complete the most basic light control.

Effect Description

After receiving commands, all RGB light beads will uniformly display the specified color, such as red, green, blue, white, or warm white. This case is suitable for verifying whether color mapping is normal.

Example Instructions

- Set all RGB lights to red
- Change the light strip to blue
- Adjust the light to warm white

6.2 Case Two: Breathing Light

Experimental Objective

Make RGB lights perform brightness gradients on a certain color to form a breathing light effect.

Effect Description

RGB lights will cycle from dark to bright, then from bright to dark. The overall effect is relatively soft, suitable for standby prompts, ambient lights, or status indicator lights.

Example Instructions

- Turn on blue breathing light
- Change the light to green breathing effect
- Slow down the breathing light speed

6.3 Case Three: Warning Flash

Experimental Objective

Make RGB lights simulate warning lights or alarm prompt effects through rapid flashing.

Effect Description

Lights can flash rapidly using red, yellow, or alternating red and blue, suitable for expressing information like "attention", "warning", "abnormal status", etc. The visual feedback will be more obvious.

Example Instructions

- Make the lights flash red
- Turn on yellow warning light
- Make RGB lights flash three times quickly

6.4 Case Four: Rainbow Marquee

Experimental Objective

Make RGB lights form continuous dynamic flowing effects to display richer light performance.

Effect Description

Light bead colors will change sequentially and flow forward, looking like rainbow flowing lights or marquee. This effect is very suitable for boot animations, display animations, or interactive light effects.

Example Instructions

- Turn on rainbow light effect
- Make the light flow like a marquee
- Turn on blue flowing light, faster speed